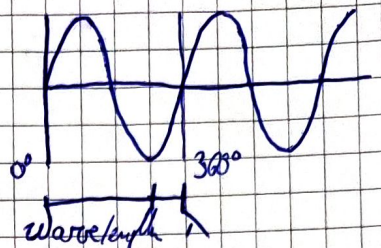
# Biophysical Characterization Week 4

## Protein Crystallography: Structure Determination & Refinement and Structure Prediction

### Structure Factor and the Phase

Reflection from a set of planes: Wave of Radiation, represented by Structure Factor, Structure Factor is a complex number

Structure Factor: $F_h = |F_h| \exp(i\phi_h)$




Intensity of a reflection is related to amplitude $|F_h|$

$$I_h \propto |F_h|^2$$

But Phases $\phi$ cannot be determined from experiment (directly)

### Phase Problem

Phases are more important than Amplitude.

↳ contain information about spatial arrangements of the atoms

Solving the Phase Problem:

→ Direct Methods: Algorithms that use probability and known chemical constraints to estimate phases

→ Molecular Replacement: Use template structures

→ Isomorphous Replacement and Anomalous Dispersion: Introducing heavy atoms into the crystal that scatter X-rays differently, providing additional data

### Data Collection

Rotation: Of the crystal → sample all angles

Cooling: Of the crystal: → reduce vibration of the atoms in the lattice

→ prevent radiation damage

# Bachelor Vorbereitung Woche 4

## Data Processing & Integration

Indexing and Integration: → Reflections are indexed (h k l)

→ Integration: Intensity for each spot (reflection)

Scaling and merging: (data reduction)

→ symmetry-related reflections are combined

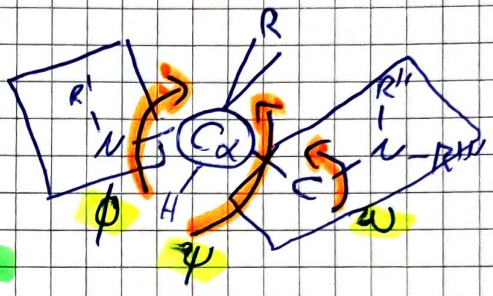
→ Merging R factor: Indicator of data quality  Rmerge = formula

→ Expectation values: < 10% ✓✓, < 30% ✓

→ Rmerge outdated → data quality: I / σ(I), $CC_{1/2}$

Software: CCP4, Phenix

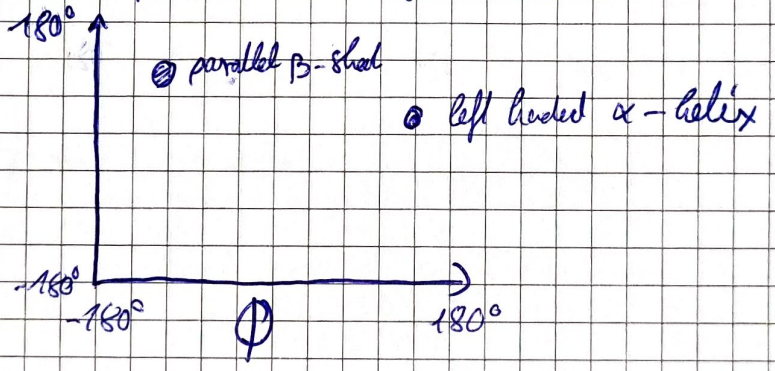
## Validation: Ramachandran Plot




→ Dihedral angles ϕ & ψ (torsion angles)

adopt typical ranges for protein structures

↳ Use these expectation values for validation




→ check if ϕ & ψ are in allowed regions

## Resolution

Resolution → the smaller the number the better, e.g.: 0,5 Å

Software to display model and electron density: Coot

Which is the bond length in tryptophan of a carbon-carbon single bond?

C - C  1,54 Å

# Bioinformatics Databases

## Protein Data Bank

Be aware of -> Validation report generated

-> Quality of structures is not checked

# Bobyland Vorbereitung – Woche 4

## Workflow

1) Protein production & purification
2) Crystallization
3) Crystal fishing
4) Data collection
5) Structure Solution & Solve the phase problem
6) Fit initial model into electron density
7) Refinement
8) Deposition
9) Application of the structural model

## Protein Folding Problem (Levinthal Paradoxon)

→ Protein fold within milliseconds into stable structure
→ Structure dictates function
→ Protein cannot sample all possible conformations
→ One Approach: Template-Based Modelling, but needs a template

## AlphaFold 2 AI

→ Predicts a protein 3D-structure from its sequence, multiple sequence alignments and potentially structural templates
→ trained also on largest structural and sequence databases

→ More Functionality ✓

Protein MPNN & Protein Structure → Sequence

→ Why Equations? → Stochastic models

# Bioinformatics Vorlesung - Woche 4

## Conclusion

-> Protein crystallography is a powerful method for determine molecular structure and has a long history and is currently still being developed

-> Protein folding problem: very difficult to predict structure based on sequence

-> Alphafold 2 & RoseMaFold: AI software for predicting structures with high success

-> NN: trained on data, such as protein structures

-> Sequence to protein structure: Alphafold 2

-> Protein structure to sequence: ProteinMPNN

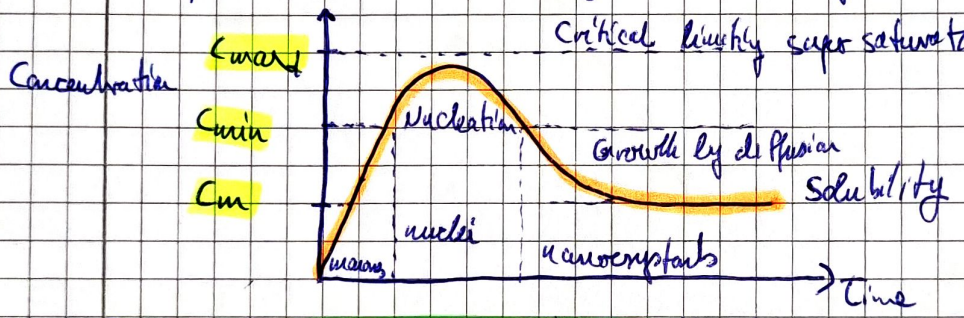
# Bidyknded Nanomaterials With S

## Nanoparticles and Nanoparticle Ligands

Nanoparticles Properties:
- → fluorescent
- → plasmonic
- → magnetic
- → catalytically active

Nanoparticles:
- → Nanoparticles are materials that exists in a size range between bulk materials (which we can see) and individual molecules (extremly small).
- → They are not atomically precise, meaning they do not consist of a specific # of atoms arranged in a perfectly defined manner.
- → Bottom-Up Synthesis: Nanoparticles ← Clusters ← Atoms
  - → Crystallization must be stopped (called growth)

### Nanoparticle growth according to the model of La Mer




Phase 1:
- Reaction precursors are initially dissolved (T=0), followed by the generation of monomers from the chemical reaction between the precursors.
- As the concentration of monomers increase above the solubility limit $C_{min}$

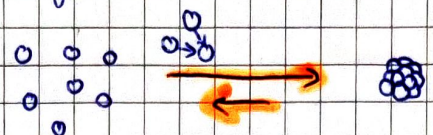
Phase 2: to the super-saturation level $C_{max}$, nuclei will be formed by aggregation and/or self-nucleation of monomers.
- Following nucleation, these nuclei serve as seeds for further growth.

Phase 3: Additional monomers in the solution attach to these nuclei, causing them to grow into larger nanoparticles. The growth of these nanoparticles continues until the monomers concentration in the solution drops below the solubility limit, preventing further growing.

# Biophysical Nanostructures

## Growth of nanoparticles: Classical Nucleation Theory

Homogeneous Nucleation (No additional ions / Reagents or surfaces):



Nucleus with radius $r_c$ (critical cluster size)

$$\Delta G = -\frac{4}{3}\pi r^3 \Delta G_V + 4\pi r^2 \gamma$$

$$\Delta G_C = \frac{16\pi (\Delta G_\Theta)^3}{(\Delta G_V)^2}$$

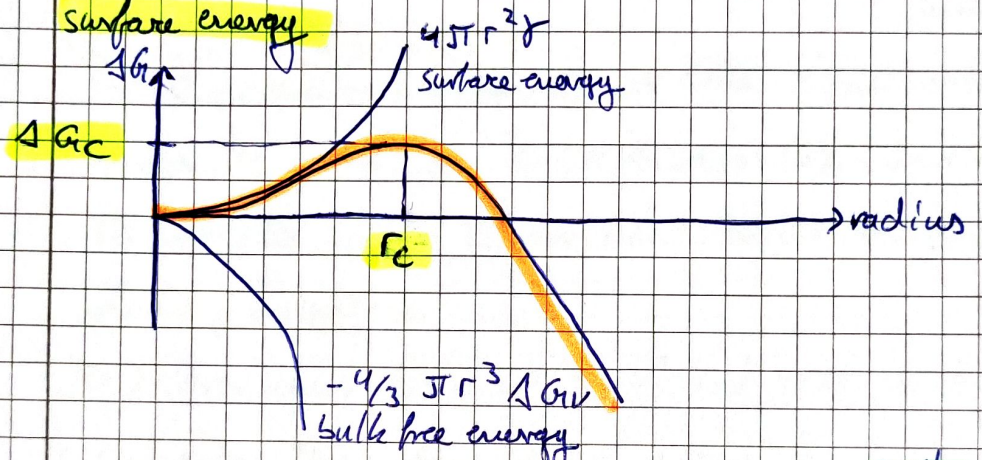
$\Delta G_C$ = Activation energy for nucleus formation with $r_c$

$\Delta G_\Theta = \gamma$ = Surface energy per Unit Area

$\Delta G_V$ = Gibbs bulk free energy per unit volume

Nucleation is favored by high T (temperature), high concentration (S) and small

surface energy




Critical Cluster Size $r_c$: The radius of a cluster of units that is at a critical size.

If a nucleus reaches this size, it has a higher probability of growing into a larger particle rather than dissolving back into the solution

Since $\rightarrow \Delta G(r_c) < 0$

# Biophysical Nanostructures Week 5

## Aggregation through coalescence or Ostwald ripening

Coalescence: Two particles merge upon collision

Driving Force: Minimization of surface energy

Ostwald Ripening: Dynamic process, exchange of surface atoms through diffusion, small particles dissolve at the expense of larger particles

Parameters:
- -> concentration, supersaturation
- -> temperature
- -> pressure
- -> surface energy

## How to control aggregation & Crystal growth?

-> Ligands: bind to the surface of nanoparticles -> Steric Stabilization & Electrostatic Stabilization

-> Electrostatic Stabilization:
controlling nanoparticle aggregation. When ligands with charged functional groups (eg: citrate) ~~are~~ bind to the nanoparticle: impart their charge. This leads to repulsion between like-charged nanoparticles, thereby preventing aggregation

eg: Gold nanoparticles can be stabilized using citrate ions

-> Aggregation upon Adding Neutral Ligands:
Introducing neutral ligands (eg: pyridine) can displace or shield charged ligands, reducing electrostatic repulsion and leading to aggregation.

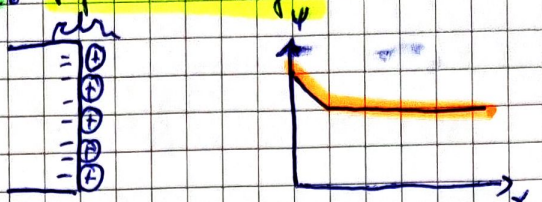
-> Crystal Growth Control:
Ligands can influence the shape and size of nanoparticles by selectively binding to certain crystal facets and inhibiting growth

# Biological Nanostructures 2024 S

## Models for charged surfaces in electrolytes

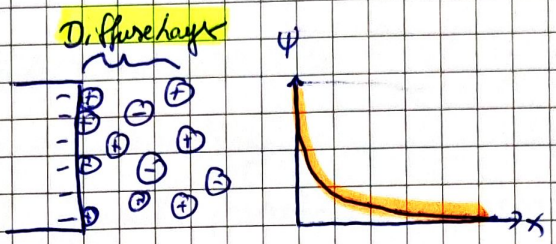
Different Perspective on Electrical Double Layer (EDL), which is the structure that forms at the interface between a charged surface and an electrolyte.

a) Helmholtz




EDL as a single rigid layer of ions, with a linear potential drop

b) Gouy - Chapman




~~Diffuse layer of ions near charged surfaces, especially~~

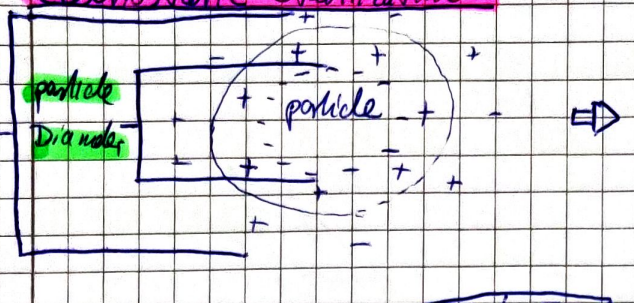
Diffuse layer of ions with an exponential potential drop, accounting for thermal motion

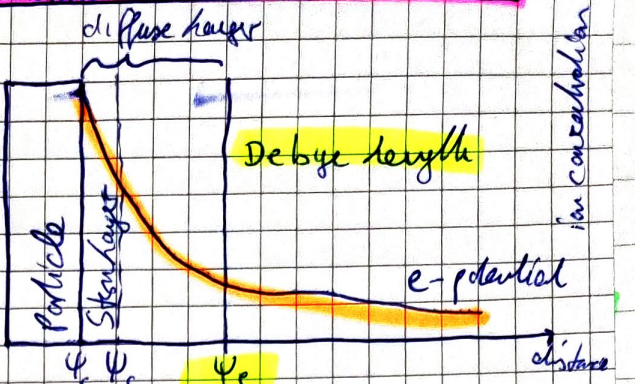
c) Stern

Combines both models, resulting in a more accurate depiction of the potential drop

# 3. Colloidal Dispersions with S

## Electrostatic Stabilization




Debye length $K^{-1} = \sqrt{\frac{\varepsilon \varepsilon_0 k_B T}{e^2 N_A \sum z_i^2 c_i}}$

↳ how far the electrostatic influence of a charged particle extends into the surrounding solution

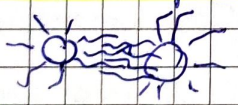
↳ range of electrostatic repulsion

## Steric Stabilization and Shielding

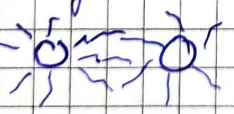
Steric Stabilization: prevents particles from aggregating through the use of large, bulky molecules that are absorbed or grafted onto the particle surfaces. (Physical Barrier)

Shielding: refers to the protective layer provided by these bulky molecules that physically prevent particles from coming close enough to each other.

↳ Repulsion through entropic and osmotic effects:

a)  Entropic Effect

As particles approach, the polymer chains on their surfaces overlap and compress, reducing their entropy. => Results in repulsive force

b)  Osmotic Effect

The overlap of polymer chains increases their local concentration, leading to an increase in osmotic pressure. => Results in repulsive force

↳ try to reduce this high concentration

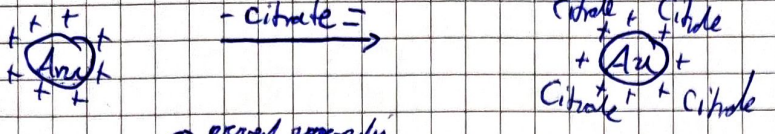
# Bioinspired Nanostructures with Au

## Synthesis of Gold Nanoparticles

General synthesis scheme for metal NP:

$M^+ + L \xrightarrow{\text{reducing agent}} M_n L_m$

Gold NP:




→ proved approach

→ Stabilization of Au with citrate ions
↳ Ligand and Reducing Agent

→ Size control & pH and ratio of starting materials

→ Turkevich Method: Slide 14

→ Optical properties of Au nanoparticles:
↳ transmitted light: red
reflected light: green

Localized surface plasmon resonance:

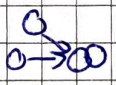
→ Oscillating field of incident light: electrons are excited to oscillate

→ Electrons: Restoring force from nucleus

→ Electric field enhanced, absorption of light at plasmon resonant frequency

## Shape control during NP synthesis

Phases of particle growth




Monomers → Nucleus → Crystal → Crystal

anisotropic growth can be controlled

→ during crystal seed formation

→ different surface energy for crystallographic faces

→ thermodynamically or kinetically controlled growth
temperature ← → pressure, concentration

→ using templates to control the shape / size

## Surface Energy and Selective Addition

From Cube to Octahedron

{100} Silver deposits on faces (100)
{111} faces stabilized (111)

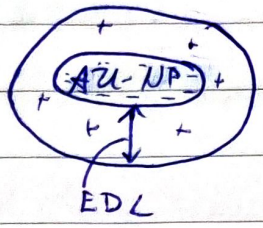
Shape control: Not understood yet (: reaction time?)

# Bimetallic Nanoparticles

## Turkevich Method: Refined synthesis of Au NPs

1. Reduction → 2. formation of gold clusters → 3. coalescence of clusters

→ 4. attachment of metal ions in the EDL → 5. reduction of the metall ions in the EDL

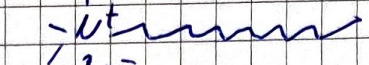


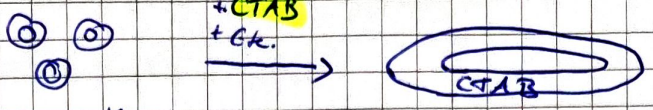

→ 6. metal atoms grow onto the particles

# Biofunctional Nanostructures WdHS

## Gold Nanorods: Seed-Mediated Growth

CTAB: Cetyltrimethylammoniumbromid



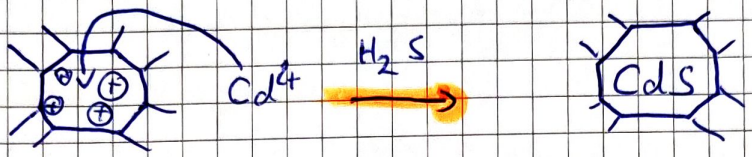



Application: Photothermal treatment

Seeds: Au NP with diameter of 3-5nm

Shape Control: Mostly empirical, parameters need to be optimized

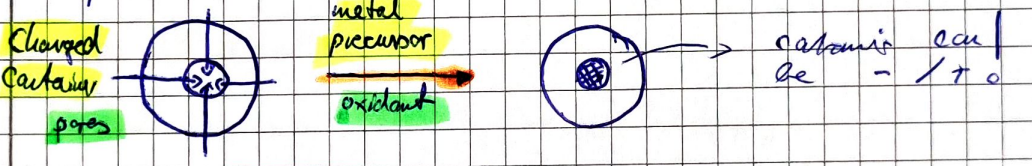
## Synthesis in porous matrices




→ Diffusion controlled process

Problem: inhomogeneous filling of pores, high polydispersity

## Nanoparticle synthesis within redox-responsive polymer carriers




## Ligands for nanoparticle synthesis

Arrested Precipitation: Particle growth is stopped

Ligands:
- prevent aggregation of particles
- prevent dissolution of particles
- provide solubility in desired solvent
- help to characterize NP
- can impart additional functionality, eg cell targeting

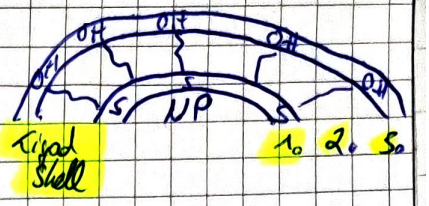
## Choices of Ligand important for later Application

For the synthesis of the seeds, $NaBH_4$ is used. For the rods, ascorbic acid is used.

$NaBH_4$ for seed generation: because it is a strong reducing agent that quickly reduces gold ions to form small nuclei for seeds

Ascorbic acid for Rod Growth: milder reducing agent allowing anisotropic growth

## Nanoparticle Ligand Shell




Ligands:

1) Head group binds to particle
2) Intramolecular region
3) Solvent-Exposed Terminal Group

Head Groups (for Gold NP):

→ weakly bound ligands such amine, CTAB
→ intermediate binding strength: citrate
→ strongly bound ligands such as thiols

Intramolecular Region:

→ forms highly packed shell, can prevent access to NP

Terminal Groups: (General)

→ charged or polar groups: water solubility (-COOH, $-NH_2$, -OH)
→ alkyl chains soluble in organic solvents

Eg for NP ligands:

PEG: high biocompatibility and stability in biological settings

Alkyl Thiols: high stabilization of NP in organic solvents

Polymers: polydentate ligands, high stability, but bulky and undefined

(large $h_s$) (short $L_s$)

⇒ Stability vs accessibility of the NP core

# Bioinspired NP Worksheet

## Ligand Exchange Reactions

$[ML_x] + L' \rightarrow [ML'_x] + L$

Phase Transfer Reaction:

→ Exploiting Solubility in Organic or Aqueous Phases

→ Phase transfer reactions utilize the different solubilities of ligands in various solvents.

→ A method used to transfer NP from one phase to another by modifying their surface properties using ligands

Janus-Type Particle:

→ Ligand Exchange on Solid support

→ NP that have two / more distinct surface regions with different chemical functionalities.

→ useful: catalysis, etc

## Protein Cargo Encapsulation with CCP

→ Thermostable

→ Cargo-loading Peptide (CCP)

## Hot Injection (TOP/TOPO)

General: Injection of precursor cools hot solution for short time, high number of nuclei formed, followed by temperature increase and particle growth

Why is it applied to carry out ligand exchange?

→ Particle stability and solubility

→ Functional & properly capping

NP: → Stabilization: electrostatic or steric

→ Shape control with ligands, reaction conditions, templates

→ Ligands important for stability & functionality

→ Ligand exchange reaction requires understanding of particle stability

# Biophysik Vorstructure Wdh 6

## Scattering Methods (SAXS, SANS, DLS)

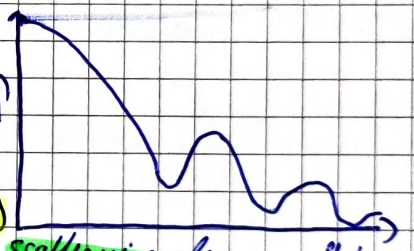
- Small Angle X-ray scattering (SAXS): Isolated particles and assemblies in solution and at surfaces
- Small Angle Neutron Scattering (SANS)
- Dynamic Light Scattering (DLS): Particles in solution and surface charge

### Advantages of scattering Methods

→ Solution-based methods (no crystal needed)
→ Relatively fast
→ Information on global average (in contrast to imaging methods)
→ Buried Structures and 3D Structures (not only on surface)
→ In-situ, Non-destructive (but radiation damage due to X-ray possible)
→ Investigation of assembly processes
→ Possibility to perform time-resolved experiments
⇒ But & Data is difficult to interpret

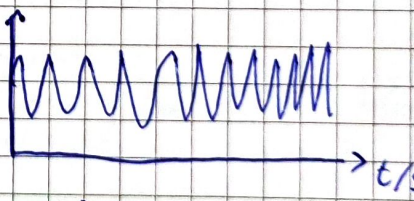
### Scattering Methods: Overview

Small Angle Scattering (SAXS & SANS):
$I(q)$ : Intensity $I$, Scattering Vector $q = \frac{4\pi}{\lambda} \sin(\frac{\theta}{2})$ (wavelength)




Key Parameter: → Conformation → Shape → Structure → Size
↳ static: snapshot of the structural info averaged over time

Dynamic Light Scattering (DLS): dynamic Properties
$I(q,t)$ : Intensity $I$, Scattering Vector $q$, Time $t$




Key Parameter → Diffusion → Rotation → Hydrodynamic Radius
↳ dynamic: time-dependance arises because DLS monitors the fluctuations in the scattered light intensity due to Brownian Motion

# Biophysikalische Methoden

## DLS

-> A typical DLS-Speckle pattern by constructive and destructive interference

-> DLS addresses those fluctuations, which are caused by the thermal motion of the scattering objects and which occur at rather small time scales (microseconds and nanoseconds)

Brownian Motion

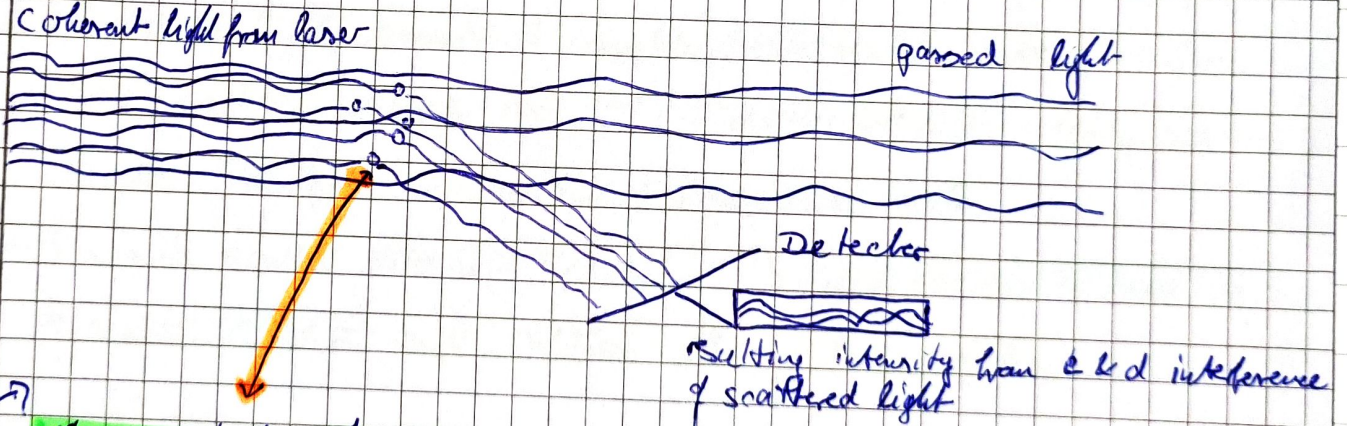
-> Relies on measuring the fluctuations in the intensity of scattered light caused by the Brownian Motion of the particle.

-> Speckle Patterns in DLS: -> monochromatic light source shd through a polarizer

When a laser beam is directed into a sample containing particles, the light scatters due to interactions with the particles. The scatted light waves interfere with each other, creating a pattern of light and dark regions known as a speckle pattern.

A) Constructive Interference: scattered light waves are in phase -> brighter spot

B) Destructive Interference: scattered light waves are out of phase -> darker spot




-> The scatted light goes through a second polarizer where it is collected by a detector and the resulting image is projected onto a screen.

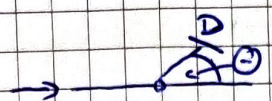
↳ Speckle Pattern

A monochromatic light source, usually a laser, is shone through a polarizer and into a sample.

# Biophysikalische Messmethoden Woche 6

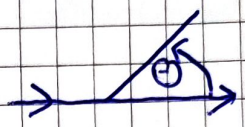
## DLS

Dynamic Light Scattering (DLS) is a technique that can be used to determine the size distribution profile of small particles or polymers in solution.




### Prinzipie of DLS:

Light scattered by a fine particle illuminated with a laser is measured with high time resolution under a defined angle Θ; the fluctuation of the scattering signal reflects the dynamics of microstructural processes such as the Brownian Motion of the particle.




### Principles and Pitfalls of DLS

→ Measuring Single Scattering Events

→ Choice of the best angle depends on the sample properties, eg: turbidity and particle size

→ Back scattering detection (eg: 173°/175°) is particularly interesting for turbid and highly concentrated samples, which contain large particles

→ Side scattering detection (eg: 90°) is recommended for weakly scattering samples, including small particles and transparent samples

→ Forward scattering detection (eg: 13°/15°) is suitable for detection of samples containing small particles with few large particles

### Errors:

The measurement of the particle size can be influenced by:

a) polydispersity – wide size distribution

b) sedimentation – gradient in count rate

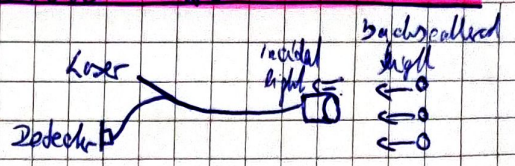
c) aggregation – increased size

d) concentration too low – Low intensity

e) concentration too high – excessive intensity

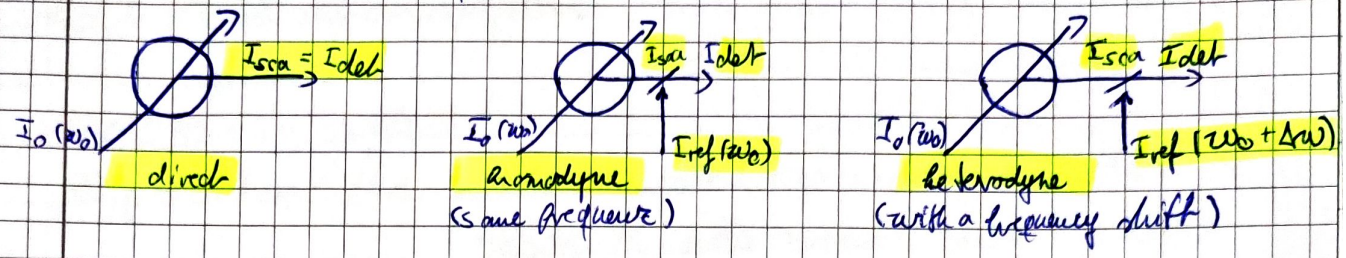
# Dynamic Light Scattering (DLS)

## DLS: Different Technical Setup




Different Setups:

Laser light of intensity $I_0$ at frequency $\omega_0$ illuminates a particle system, which scatters a part of the light energy in all directions; either unprocessed or modulated by superposing with a reference beam of same or different frequency as the incident light.




### Autocorrelation Function

The dynamic information of the particles is derived from the autocorrelation of the intensity trace recorded during the experiment.

The second order autocorrelation curve is generated from the intensity trace as follows:

$$g^2(\tau) = \langle I_{det}(t) \cdot I_{det}(t+\tau) \rangle / \langle I_{det}(t) \rangle^2$$

where $g^2(\tau)$ is the autocorrelation function at a delay time $\tau$ and $I$ is the intensity.

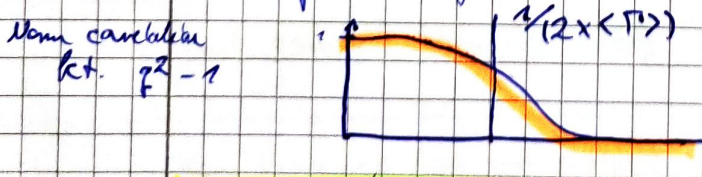
$$\Gamma = s^2 D_p$$

Decay rate is directly proportional the particle diffusion coefficient $D_p$

The factor of proportionality is the scattering vector $s$:

$$s = |\mathbf{s}| = \frac{4\pi n_f}{\lambda_0} \sin\left(\frac{\Theta}{2}\right)$$

with $n_f$ denoting the refractive index of the fluid.




The exponential decay is related to the motion of the particles, specifically to the diffusion coefficient.

# Biohybrid Nanostructures Wdh 6

## Stokes - Einstein Sutherland Equation

Particle diffusion coefficient $D_p$, which is inversely proportional to particle size. $D_p = \frac{k_B T}{3\pi\eta x_{h,t}}$

$x_{h,t}$: translational hydrodynamic diameter
$\eta$: dynamic viscosity
$k_B$: Boltzmann's constant
$D_p$: Particle Diffusion Coefficient: measures how fast the particle diffuse through the fluid

## Dispersity of the sample

$Đ_M = M_w / M_n$, where $M_w$ is the mass-average molecular weight, $M_n$ is the number-average molecular weight

## Polydispersity Index (PDI) from DLS

In DLS the size distribution of molecules or particles is the property of interest.
Here, the distribution describes how much material there is present of the different size "slices"

$$\text{PDI from DLS} = \left(\frac{\text{Standard Derviation}}{\text{Mean}}\right)^2$$

PDI from GPC $= M_w/M_n$

If PDI > 0.5, the distribution is rather broad, and the reliability of data is frequently reduced. eg: Paderin Catani: PDI = 0,031 ✓
Boiled UP: PDI = 0,519 ✓

## Signal Quality for Backward Scattering Data

poor quality: noise at large $\tau$, amplitude partly > 1, non-exponential decay, large variation of results
good quality: exponential decay, high repeatability, amplitude > 0.8

Main Reasons: (for bad quality)
Particle Concentration is either too high or too low.
Dust particles or gas bubbles can also reduce signal quality.

## Zeta ($\zeta$) potential and its significance

Zeta potential is a function of the surface charge and reflects the electrical potential at the slipping plane.
This plane is the interface which separates mobile fluid from fluid that remains attached to the surface.
The Zeta potential is an important and readily measurable indicator of the stability of colloidal dispersions. → oil-water dispersions

- $-30\ mV < \zeta > 30\ mV$: stable dispersion because particles repel each other
- $\zeta \Rightarrow 0\ mV$ (Isoelectric Point): stability decreases → mixture of two phases
- depends on pH, ionic strength, additives, ligands, etc.

↳ Electrophoretic light scattering (ELS)

→ An E-field is applied across the solution using electrodes positioned on either side of the cell
→ The E-field causes the charged particles in the solution to move. Positively charged particles will move towards the negative electrode, while negatively charged particles will move towards the positive electrode.
→ Light scattering measurement
→ The scattered light is analyzed to determine the Doppler Shift caused by the movement of the particles.
This Doppler Shift is related to the electrophoretic mobility of the particles

⇒ Electrophoretic mobility is measured through ELS:

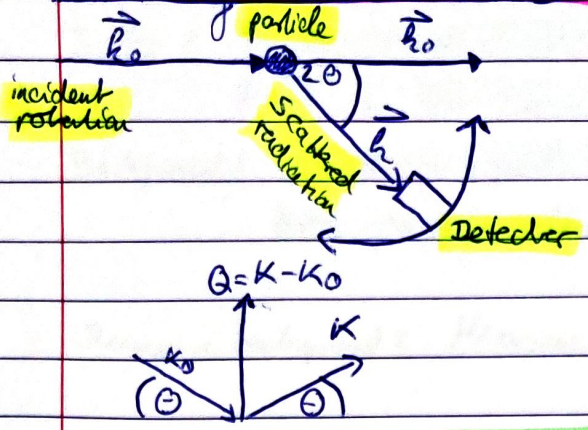
→ electric field is applied
→ mobility of the scattered particles is measured

→ i.e. the constant particle velocity, the electrophoretic mobility is expressed by the Henry Equation:

$$\mu = \frac{\varepsilon \cdot \zeta}{\eta}$$

# Biophysical Characterization Wdh 6

## Scattering Methods: SAXS




For elastic scattering:

$$|\vec{k_0}| = \frac{2\pi}{\lambda} = |\vec{h}|$$

scattering vector:

$$\vec{q} = \vec{k_0} - \vec{h}$$

$$|\vec{q}| = \frac{4\pi}{\lambda} \sin\Theta$$

Dimension of $q$ = length$^{-1}$ eg $nm^{-1}$

Small Angle X-rays: powerful tool to study materials at the nm–µm scale.

Provides Info About: size, shape and distribution of particles

Method:

A collimated beam of X rays is directed at a sample, and the scattered radiation is measured at small angles. This scattering occurs due to the interaction of the X ray beam with the electron clouds of the atoms.

Key Factors: Incident Wave Vector $\vec{k_0}$, Scattered Wave Vector $\vec{h}$, scattering vector $\vec{q}$

Dimension of $\vec{q}$:

[$nm^{-1}$], this inverse length scale is crucial for interpreting the size and shape of structures in the sample. Relation between the scattering vector and the spatial dimensions of the sample structures is inverse.

↳ Smaller Structure ~ larger $|\vec{q}|$

$I(q)$: Scattered Intensity

$P(q)$: form factor; depends on size and shape, dominant for high q-values → Spheres, Rods, Bilayers

$S(q)$: structure factor; interaction between particles, depends on the concentration

$$I(q) \propto P(q) \cdot S(q)$$

Structure factor ≈ the same as in crystallography!

Biophysikalische Messmethoden WS 16/17 #6

## SAXS

Buffer: nanoparticles are suspended in a liquid medium called a buffer. Necessary for their stability and function

Background: Scattered signal that comes from the buffer and other non-particle elements.

! Remove Background: Measure the Total Scattering − Measure the Buffer alone

### Guinier: Mean particle size determination with SAXS

For the Guinier approximation, the scattered intensity is plotted as function of $q^2$ (Guinier Plot), and the radius of gyration can be obtained from the slope at very low q-values (Guinier Region)

$$I(\vec{q}) \approx I(0) \exp(-\tfrac{1}{3} \cdot (\vec{q} \cdot R_g)^2)$$

$I(0)$ → overall measure of particle size; $R_g$ → radius

### Model Fitting: Mean particle size determination → detailed

$P(q)$ form factor: depends on size and shape, dominant for high q-values

$S(q)$ structure factor: depends on the arrangement of the scattering objects and simplifies for example in a dilute dispersion where the interaction between the particles and multiple scattering can be neglected

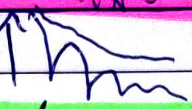
$$I(q) = P(q) \cdot S(q) \xrightarrow{S(q)=1} I(q) = P(q)$$

Form Factor Model: eg: spheres, cylinders, ellipsoides and more

↳ Fit the theoretical $P(q)$ model to the experimental data $I(q)$

# Bobylside Nanobuilding Wk #6

## SAXS: Polydispersity



Polydispersity: variation in size and shape among particles within a sample

distribution of particle size and shapes

→ range of scattering contributions → oscillation amplitudes decrease and the peak become less distinct

Monodisperse: clear minima at certain q-values

→ minimas smear out

The broader the distribution the pronounced the smearing effects leading to loss of detailed structural information.

Problem: Loss of Detail, Difficulty in Model Fitting, Ambiguity in Size Determination, Interpretation Challenges

What helps?

- Narrowing the distribution: reduce polydispersity
- Polydispersity Models, Regularization Techniques

Guinier

1) The periodicity of the oscillation on the q-axes is proportional to the mean diameter
2) Size distribution from intensity pattern

## Biological SAXS

→ Carried out in solution, no crystal required

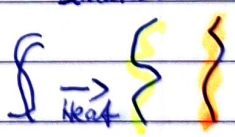
→ Averaging over all shapes and sizes

→ Obtain the envelope of protein

→ Dynamics in Solution

→ Low Resolution

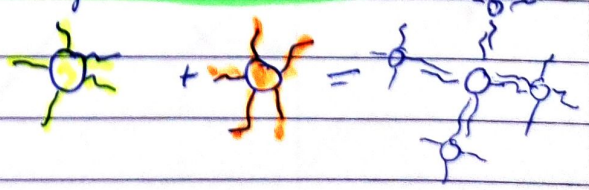
denaturation




→ DNA as ligands for NP assembly

↳ NPs can be functionalized with single stranded DNA

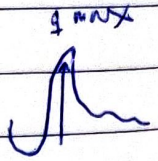
↳ assembly / aggregation of NP



# Biohybride Nanostructure Woche 6

## SAXS: Structure Factor

Structure factor S(q): interaction between particles in space



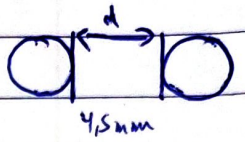

-> high particle concentration

-> particle assembly / aggregation $q_{max} \cdot d \approx 2\pi$

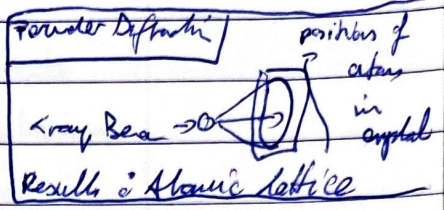
$d \approx \frac{2\pi}{q_{max}} - 2 R_g = 4{,}5\ nm$

↳ factor for sphere

↳ radius of gyration




Small angle X-ray diffraction SAXD




-> Ordering on the nanoscale

disordered some ordering oriented (textured) single crystal

<- Disorder ——— more order ->

-> SAXS pattern reveal nanoscopic order

-> Powder diffraction: Position of atoms

-> Now: Not atoms are the building blocks of lattice, but NP

-> NP: ordering into superlattice interaction with X-rays

Result: Position of NP ↳ Periodic

↳ Analyze Diffraction Peaks

## Assembly of NP with protein containers

Redesigned ferritin containers:

-> filled with metal oxide NPs

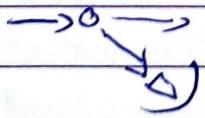
-> assembly into highly ordered structures

-> highly ordered lattice

-> large domains (µm)

-> can be stabilized

SAXS vs SAXD



measures electron density variations at small angles

measures diffraction pattern formed by X-rays

# Biophysical Characterization WoA 6

## Neutron Scattering

Generation:
- → Nuclear reaction, e.g. decay of uranium emits neutrons
- → Spallation sources: target is bombarded and emits

Advantages over SAXS:
- → Sensitivity for light elements
- → possibility of isotope labelling
- → the strong scattering by magnetic core moments

Application: Contrast variation & labelling with deuterium (solvent effects, magnetic characterization)

Neutron scattering length:

The scattering length density and is a measure of the strength of the interaction of a neutron wave with a given nucleus, eg: H = -0,37 ↓ 180°

↳ feature is important for the technique of contrast variation

D = 0,667

SANS contrast variation and biological science:

To use contrast variation, different components of a system must scatter differently. This can be based on inherent scattering differences, e.g. DNA vs Protein, or arise from differentially labeled components.

↳ Characterization of ligand shell with SANS contrast variation

→ NPs with different shells

## Conclusion

- → DLS: Hydrodynamic diameter (potential issues with reproducibility and polydisp)
- → Zeta Potential: Surface charge of particles
- → SAXS can give information about size, shape, concentration and assembly of NPs
- → Small angle X-ray diffraction: Information on superlattice of NPs
- → Neutron Scattering: Contrast variation: small angle neutron scattering (SANS) combined with deuterium labelling makes it possible to define the position of specific components within a complex SAXS provides more precise data on the overall shape while